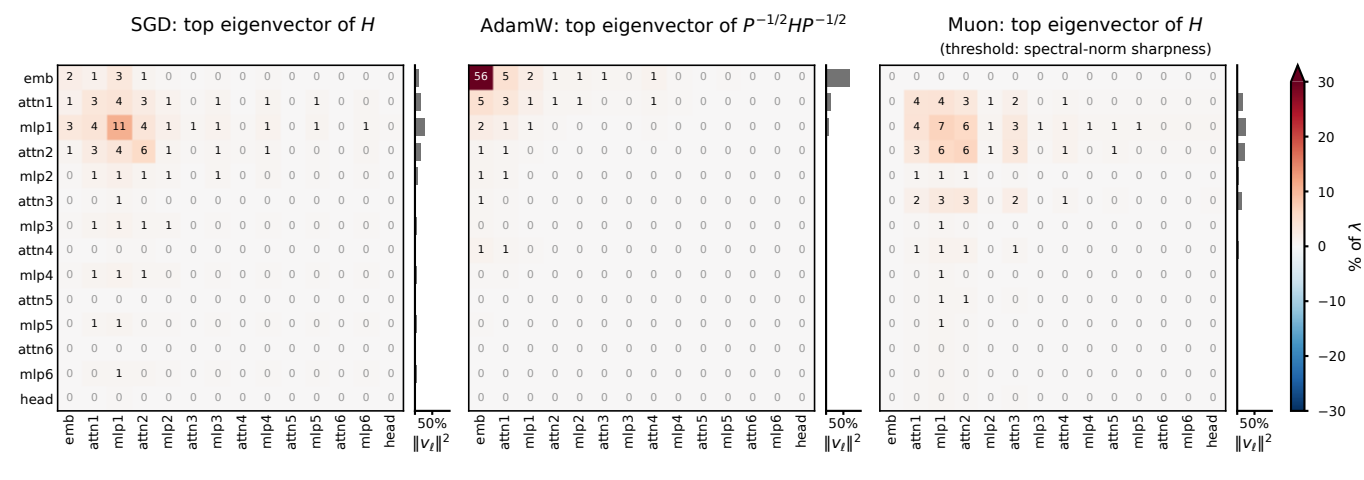
SGD: top eigenvector of H
AdamW: top eigenvector of $P^{-1/2}HP^{-1/2}$
Muon: top eigenvector of H
(threshold: spectral-norm sharpness)
emb
attn1
mlp1
attn2
mlp2
attn3
mlp3
attn4
mlp4
attn5
mlp5
attn6
mlp6
head
50%
$\|v_t\|^2$
% of λ
30
20
10
0
−10
−20
−30